

## 1. Problem 1 — Convert Celsius to Fahrenheit

Requirement: Write a function that takes a temperature in Celsius and converts it to Fahrenheit using:

$$F = (C \times 9/5) + 32$$

### Source Code

```
function celsiusToFahrenheit(celsius) {  
  return (celsius * 9 / 5) + 32;  
}  
  
// Test  
let celsius = 25;  
let fahrenheit = celsiusToFahrenheit(celsius);  
  
console.log("Celsius:", celsius);  
console.log("Fahrenheit:", fahrenheit);
```

### Sample Input

Celsius = 25°C

### Output

Fahrenheit = 77°F



The screenshot shows a code editor with a dark theme. The code is as follows:

```
1 function celsiusToFahrenheit(celsius) {  
2   return (celsius * 9 / 5) + 32;  
3 }  
4  
5 // Test  
6 let celsius = 25;  
7 let fahrenheit = celsiusToFahrenheit(celsius);  
8  
9 console.log("Celsius:", celsius);  
10 console.log("Fahrenheit:", fahrenheit);
```

Below the code editor, the 'Debug Console' tab is active, showing the output of the code:

```
C:\Program Files\nodejs\node.exe .\DIU DBMS Project\backend\routes\function celsiusToFahrenheit(celsius) (.js  
Celsius: 25  
Fahrenheit: 77
```

### Explanation

The function receives a Celsius value as the parameter `celsius`. It applies the given formula, multiplies the value by 9/5, adds 32, and returns the Fahrenheit temperature. For 25°C, the result is 77°F.

## 2. Problem 2 — Find the Factorial of a Number

Requirement: Write a function that calculates the factorial of a given non-negative integer. For example,  $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ .

### Source Code

```
function factorial(n) {  
  if (!Number.isInteger(n) || n < 0) {  
    throw new Error("Factorial is defined for non-negative integers only.");  
  }  
  
  let result = 1;  
  
  for (let i = 2; i <= n; i++) {  
    result *= i;  
  }  
  
  return result;  
}  
  
console.log(factorial(5));
```

### Sample Input

Number = 5

### Output

Factorial = 120



The screenshot shows a code editor with a dark theme. The top part displays the JavaScript code for the factorial function, which includes input validation and a loop to calculate the factorial. The bottom part shows the output of the code, which is the number 120. The code is as follows:

```
1 function factorial(n) {  
2   if (!Number.isInteger(n) || n < 0) {  
3     throw new Error("Factorial is defined for non-negative integers only.");  
4   }  
5  
6   let result = 1;  
7  
8   for (let i = 2; i <= n; i++) {  
9     result *= i;  
10  }  
11  
12  return result;  
13 }  
14  
15 console.log(factorial(5));
```

The output of the code is 120, displayed in the console at the bottom of the editor.

## Explanation

The function first validates that the input is a non-negative integer. The variable result starts at 1. The for loop multiplies result by every integer from 2 through n. Therefore, for n = 5, the calculation becomes  $1 \times 2 \times 3 \times 4 \times 5 = 120$ . The same function also correctly handles 0!, which is 1.

## 3. Problem 3 — Check for Palindrome

Requirement: Write a function that checks whether a given word reads the same backward as forward and returns true or false.

### Source Code

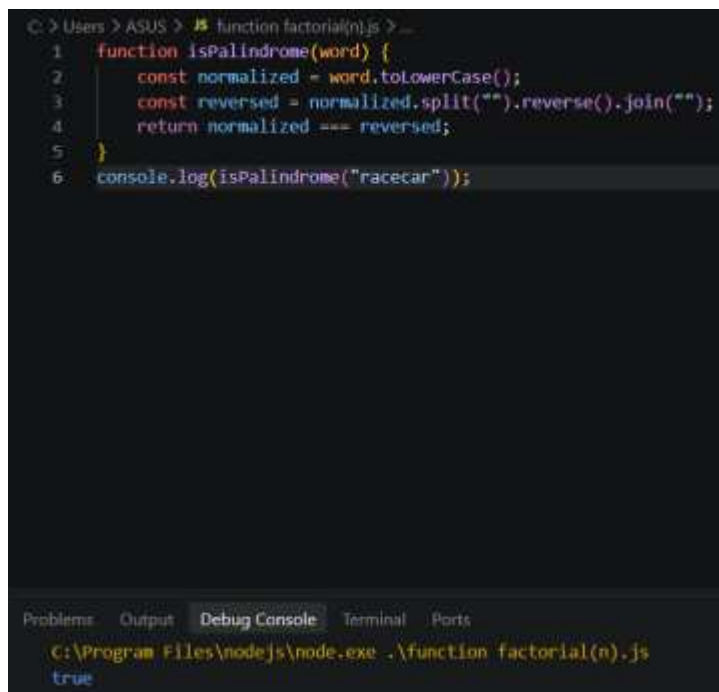
```
function isPalindrome(word) {  
  const normalized = word.toLowerCase();  
  const reversed = normalized.split("").reverse().join("");  
  return normalized === reversed;  
}  
  
console.log(isPalindrome("racecar"));
```

### Sample Input

Word = "racecar"

### Output

True



The screenshot shows a code editor with a dark theme. The code defines a function `isPalindrome` that takes a word, converts it to lowercase, splits it into an array of characters, reverses the array, and joins it back into a string. It then compares the original string with the reversed string. The function is called with the argument "racecar", and the result is logged to the console. The output in the console is `true`.

```
C:\Users\ASUS> .\function factorial(n).js  
1 function isPalindrome(word) {  
2   const normalized = word.toLowerCase();  
3   const reversed = normalized.split("").reverse().join("");  
4   return normalized === reversed;  
5 }  
6 console.log(isPalindrome("racecar"));
```

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C:\Program Files\nodejs\node.exe .\function factorial(n).js  
true

## Explanation

The word is converted to lowercase so that capitalization does not affect the comparison. `split("")` converts the word into an array of characters, `reverse()` reverses that array, and `join("")` creates the reversed string again. Finally, the original normalized word is compared with the reversed word. Since `racecar` is identical in both directions, the function returns `true`.

## 4. Problem 4 — Sum of Array Elements

Requirement: Write a function that takes an array of numbers and returns the sum of all elements in the array.

### Source Code

```
function sumArray(numbers) {  
  return numbers.reduce((sum, number) => sum + number, 0);  
}  
  
console.log(sumArray([1, 2, 3, 4, 5]));
```

### Sample Input

Array = [1, 2, 3, 4, 5]

### Output

Sum = 15

A screenshot of a code editor with a dark theme. The editor shows a JavaScript file named 'function factorial(n).js'. The code defines a function 'sumArray' that uses 'reduce' to calculate the sum of an array. Below the function definition, there is a call to 'console.log' with an array of numbers [1, 2, 3, 4, 5]. The editor's interface includes a 'Debug Console' tab at the bottom, which shows the output of the code: '15'. The file path in the editor is 'C:\Program Files\nodejs\node.exe .\function factorial(n).js'.

## Explanation

The `reduce()` method processes every element of the array and accumulates the total. The initial value of the accumulator is 0. The values are added as  $0 + 10 + 20 + 30 + 40 + 50$ , which produces 150.

## 5. Problem 5 — FizzBuzz (Basic)

Requirement: Write a loop from 1 to 15 that prints "Fizz" for numbers divisible by 3, "Buzz" for numbers divisible by 5, "FizzBuzz" for numbers divisible by both, and the number itself otherwise.

### Source Code

```
function fizzBuzz() {  
  for (let i = 1; i <= 15; i++) {  
    if (i % 3 === 0 && i % 5 === 0) {  
      console.log("FizzBuzz");  
    } else if (i % 3 === 0) {  
      console.log("Fizz");  
    } else if (i % 5 === 0) {  
      console.log("Buzz");  
    } else {  
      console.log(i);  
    }  
  }  
}
```

### Input

Range = 1 to 15

### Output

1  
2  
Fizz  
4  
Buzz  
Fizz  
7  
8  
Fizz  
Buzz  
11  
Fizz  
13  
14  
FizzBuzz

```
> Users > ASUS > JS function factorial(n).js > ...
1  function fizzBuzz() {
2      for (let i = 1; i <= 15; i++) {
3          if (i % 3 === 0 && i % 5 === 0) {
4              console.log("FizzBuzz");
5          } else if (i % 3 === 0) {
6              console.log("Fizz");
7          } else if (i % 5 === 0) {
8              console.log("Buzz");
9          } else {
10             console.log(i);
11         }
12     }
13 }
14
15 fizzBuzz();
```

```
C:\Program Files\nodejs\node.exe .\function factorial(n).js
```

```
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
```

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```
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
```

## Explanation

A for loop runs from 1 through 15. The modulo operator (%) checks divisibility. The condition for divisibility by both 3 and 5 is checked first so that 15 produces FizzBuzz. If the number is divisible only by 3, it prints Fizz; if divisible only by 5, it prints Buzz; otherwise, the number itself is printed.

